



Effect of *Cucumber mosaic virus* on the Content of Some Melon Genotypes of Some Mineral Elements and Plant Hormones

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Abstract:

This study was conducted to determine the effect of *cucumber mosaic virus* (CMV) infection on the levels of some mineral elements and plant hormones in some melon genotypes. All experiments mentioned in this research were conducted in the Laboratory of plant virology and Greenhouses belonging to the Plant Protection Department/ College of the Agriculture/ University of Kerbala. PCR amplification using the complementary DNA (cDNA), synthesized from RNA isolated from an infected plant, produced a 650bp PCR product. A BLAST search using the nucleotide sequence of the PCR product demonstrated that this CMV isolate was

previously registered at the National Centre for Biotechnology Information (NCBI) with a similarity of 100% with some CMV isolates identified in Hungary (AJ517802), Australia (U22821), and Slovenia (OL142046). It was also found that all melon genotypes evaluated in this study were susceptible to CMV, and the genotypes HA-144 and HA-609 were the most susceptible to CMV among the other tested genotypes. Results also showed that the CMV infection significantly reduced the levels of some minerals (calcium, magnesium, and manganese), and the genotypes HA-144, GM-3034, and GA-1534 were the most affected by the virus; whereas the mineral elements sodium and potassium were significantly increased in their levels in infected plants, especially in the genotype HA-609. The CMV infection also clearly reduced the level of the hormone gibberellin in genotypes and the GA-1534 genotype was the most impacted and significantly different from their levels in the uninfected plants. However, the CMV-infected genotypes showed a significant increase in the level of the hormones cytokinin.

Keywords: *Cucumber mosaic virus*, minerals, genotypes, plant infection, melon.

Introduction

The melon plant (*Cucumis melo* L.), which belongs to the *Cucurbitaceae* family, is one of the most important and commonly grown vegetable crops in the world due to the nutritional and medicinal value of its fruits (Manchali et al.,

2021). Melon plants are produced in rainy and cold climates that extend the growing season, hot and dry weather that allows the melon plants to flourish and produce a product of high quality, aroma, and flavor (Ortali and Onarinde, 2021). The world's production is about 28.3 million tons of melon on average, with China



producing the first producer (15 million tons), followed by Turkey, Iran, the United States of America, and Spain (FAOSTAT, 2015). In the Arab world, Egypt is the most productive country.

Many diseases, including those caused by plant viruses such as *Cucumber mosaic virus* (CMV) can infect the melon plants. CMV is one of the factors that contributes to a decrease in yield in quantity and quality (Kimura et al., 2020). Infection can also cause plant failure in growth and production, especially in cases of early infestations of plant life (Hull, 2002). More than 1287 plant species from 100 plant families are infected by CMV (Bhardwaj and Purohit, 2021). Aphids can also spread the virus in a non-persistent manner and *Aphis gossypii* and *Myzus persicae* are the most effective aphid species in plant virus transmission (Tungadi et al., 2021). It was found from previous experiments, CMV caused a cucumber crop loss of 100% when plants infected plants at the seedling stage and 66.7% at the three-leaf stage, but only a loss of 22.2% when plants infected with the age of six true leaves (El-Gendi et al., 2022).

The polymerase chain reaction (PCR) has been identified as one of the most reliable methods for identifying plant viruses, such as *Tomato yellow leaf curl virus* (TYLCV), potato virus Y, and *cucumber mosaic virus* (CMV) (Min et al., 2020). This study aimed to determine the impact of CMV infection on the levels of certain mineral elements (calcium, magnesium, manganese, sodium, and potassium) and plant hormones (hormone gibberellin, cytokinin) in some melon genotypes.

Materials and Methods

The Source of the Virus Isolate

Leaf samples were collected from cucumber plants showing disease symptoms typical of mottling, deformation, and stunting during the agricultural season 2020–2021. These samples were collected from several greenhouses located in the Jableh district in the Babylon governorate. The samples were brought to the laboratory of Plant Virology in the Plant Protection

Department/ College of the Agriculture/ University of Karbala and used for the mechanical inoculation of new melon plants to maintain the source of the virus.

Plant Nucleic Acid (RNA) Extraction and cDNA Synthesis

Total RNA was extracted from the infected melon plants using the Taiwan FavorPrep™ Plant total RNA mini kit (Cat. No: FAPGK001). cDNA was constructed using the Reverse Transcription kit (Cat. No. LOT). - 0000487645, Promega, USA).

PCR Amplification to Amplify the CMV-Partial Coat Protein

PCR amplification using the synthesized cDNA was performed using the following parameters: initial denaturation at 94°C for 1 min followed by 35 cycles each consisting of final denaturation at 94 °C for 30 sec, annealing temperature at 55 °C for 30 sec, initial extension for 1 min, and final extension at 72°C for 5 min. PCR-amplified products were electrophoretically separated on a 1% agarose gel for 140 min at 80 V, 400 mA and visualized with ethidium bromide staining under UV illumination and images were captured using Vilber Lourmat, Taiwan gel documentation system.

The PCR-amplified products were gel-purified and sent along with the primer pair to the Macrogen DNA sequencing service in Korea. PCR products were directly sequenced in both directions using the respective forward and reverse primers. The obtained nucleotide sequences were then aligned and compared with the sequences belonged to the other fungal isolates and previously published in NCBI database using the Basic Local Alignment Search Tool (BLAST).

Testing the Response of Melon Genotypes to CMV

Ten different melon genotypes, provided by the Green Meadows Company/ Babylon Governorate, were tested against CMV (Table 1). Seeds of these genotypes were planted separately in cork plates containing sterilized peat moss in an insect-free greenhouse. The

plants were then transplanted after 15 days of germination to the soil in the greenhouse. Three replicates, which were placed randomly over the

experiment site, were transplanted with each genotype.

Table 1. The Melon Genotypes Used in this Study to Determine Their Response to CMV

Sequence	Genotype	Origin	The Producing company
1	AH-2018	Holland	Gold seeds
2	HA-144	Holland	Gold seeds
3	HA-265	Holland	Gold seeds
4	HA-17NM-106	Holland	Gold seeds
5	HA-609	Holland	Gold seeds
6	GA-1534	China	Ananas melon
7	HA-2011	Holland	Gold seeds
8	HA-17ME-214	Holland	Gold seeds
9	GM-3034	China	Ananas melon
10	GEDIZ F1	Turkey	Yuksel tohum

These genotypes' (4-6 true leaf stage) were mechanically inoculation with the virus and a control treatment (negative control) was performed by inoculating other plants only with phosphate buffer solution. These plants were monitored periodically to record any appearance and development of any disease symptoms. All experimental plants were evaluated by reverse transcription polymerase chain reaction (RT-PCR) after 30 days of inoculation, and the severity of infection was then determined using the formula previously described by Murphy et al (2003).

The Effect of CMV on the Content of Melon Fruits of Some Minerals and Hormones

The level of some minerals (calcium, sodium, and potassium) in the fruit samples taken from the genotypes were analyzed using the methods described by Jones (1984); whereas the analysis of magnesium and manganese was analyzed using the method described by Page and others (1982). The effect of CMV on gibberellin, AND cytokinin in the fruits was also done using the procedure described by Plank and Wanger (1986).

Statistical Analyses

Descriptive statistics have used Frequency, percentage tables mean and standard deviation). F test was used to compare means between three or more independent groups. Level of significance of ≤ 0.05 was considered as significant difference. Least significant difference (LSD) was used to find differences between groups (Al-fahham, 2018).

Results and Discussion

Molecular Diagnosis of CMV

A 650bp PCR product was amplified from the CMV coat protein. Analysis of the nucleotide sequence, obtained from the isolated virus, using the BLAST program indicated that this isolate belongs to CMV that had 100% identity with those isolates previously identified in Hungary (AJ517802), Australia (U22821), and Slovenia (OL142046), have a 100% similarity rate (Figure 1).

In this study, CMV was identified using polymerase chain reaction (PCR) technology (AL-Abedy et al., 2018; Al-Fadhal et al., 2018; Al-Salami et al., 2019; Al-Fadhal et al., 2019; Torrance et al., 2020; AL-Abedy et al., 2021). In a previous study, Al-Yasiri (2016) successfully

used the PCR technique to identify CMV that was isolated from cucumbers. The primers employed in this investigation were highly

effective and previously used by Chen (2003), and Asad et al (2019) in identifying CMV.

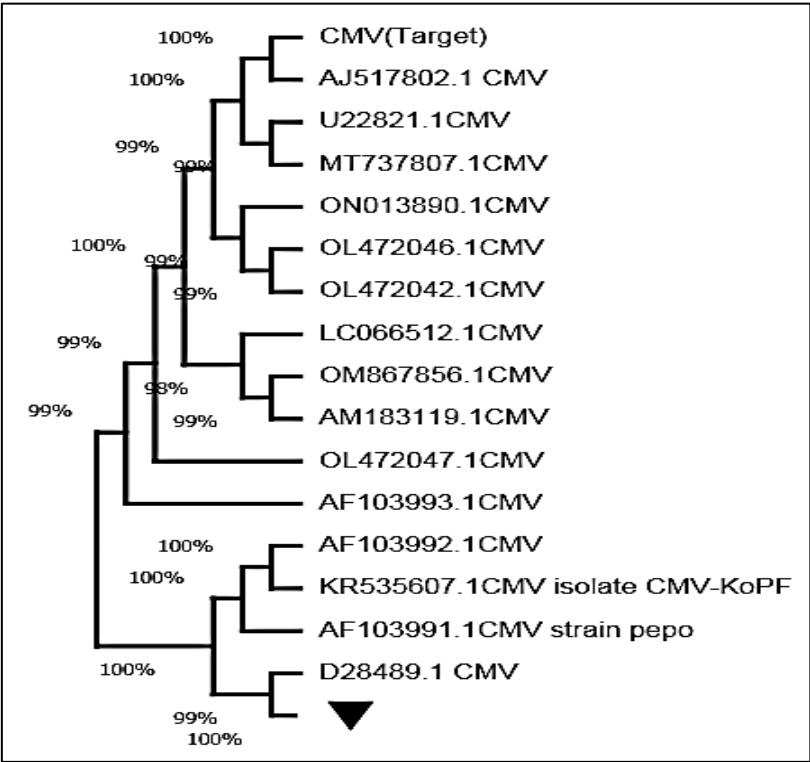


Figure 1. The Neighbor-Joining Tree Shows the Genetic Relationship of the CMV Isolate Identified in this Study and the Other Isolates Previously Registered at NCBI

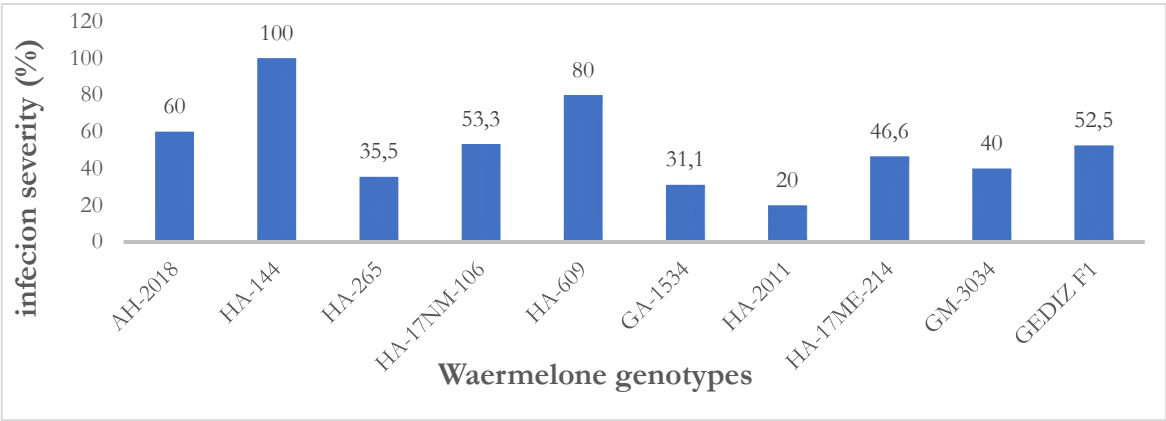


Figure .2. Severity of Infection with CMV in Melon Genotypes Tested in this Study

Severity of Infection in Genotypes

As shown in Figure 2, the results demonstrated that all melon genotypes were fully susceptible to CMV and these genotypes were varied in the

appearance and severity of the disease symptoms. Among these genotypes, it was revealed that the genotypes HA-144 and HA-609 are the most susceptible to CMV with infection severity values 80 and 100%,

respectively. The severity of infection was between 52.5 and 60% for the genotypes AH-2018, HA-17NM-106, HA-17ME-214, and GEDIZF1; whereas the genotypes HA-265, GA-1534, and HA-2011 displayed sensitivity to the virus.

The genetic variation of those genotypes and their contents of chemicals and resistance to the virus may be the cause of variation of these genotypes in their severity to CMV. These findings are in agreement with many researchers who found that different tomato genotypes are fully susceptible *Tomato yellow leaf curl virus* (TYLCV) (Karim, 2016; Abbas and AL-Abedy, 2017).

The Effect of CMV on the Mineral Content of Melon Plant Fruits

Calcium

It was discovered that calcium content in melon plant genotypes was significantly decreased

when infected with CMV at a rate of 43.00 mmol L⁻¹, which was significantly different from the normal level in the control plants, which averaged 46.54 mmol L⁻¹. The genotype HA-17NM-106 had the highest amount of this element, with an average of 46.42 mmol L⁻¹, while the genotype HA-144 had the lowest level of calcium content among the other genotypes with an average of 41.32 mmol L⁻¹. The calcium level of the remaining genotypes ranged from 46.4 to 41.32 mmol (Table 2). The interaction between genotypes that were infected and those that were not revealed that the HA-144 genotype under the influence of virus infection had the lowest calcium content of 40.57 mmol. L⁻¹ and its difference with the content of plants that were not infected with the virus, which gave an average of 42.07 mmol. L⁻¹, while the highest level of calcium (44.40 mmol. L⁻¹) was in the genotype HA-17NM-106, which was significantly different from the concentration of uninfected plants.

Table 2. The Effect of CMV on the Genotypes of Calcium Content in Melon Plant Fruits

Genotype	Calcium		Mean
	Non-infected plants	Infected plants	
AH-2018	48.67	42.47	45.57
HA-144	42.07	40.57	41.32
HA-265	46.00	43.80	44.90
HA-17NM-106	48.43	44.40	46.42
HA-609	45.03	44.03	44.53
GA-1534	47.47	43.93	45.70
HA-2011	49.43	42.33	45.88
HA-17ME-214	46.73	43.57	45.15
GM-3034	45.97	43.63	44.80
GEDIZ F1	45.57	41.30	43.43
Mean	46.54	43.00	
L.S.D _(0.05)	Genotypes	Infection	Interaction
	1.23	0.55	1.74

Magnesium

The results showed that the magnesium content in the genotypes infected with CMV reduced with a rate of 2.15 mmol L⁻¹, compared to its normal level in the non-infected plants that was 2.90 mmol L⁻¹. The genotype GM-3034 had the highest level of magnesium with an average of 3.22 mmol L⁻¹, whereas the genotype HA-609 had the lowest level of magnesium among the

other genotypes with an average of 1.67 mmol L⁻¹. The level of the same element in the remaining genotypes ranged between 3.22 and 1.67 mmol L⁻¹ (Table 3).

The interaction between the virus-infected and uninfected genotypes revealed that the virus-infected HA-609 genotype had the lowest magnesium content of 2.51 mmol. L⁻¹, which differed significantly from its content in plants

not infected with the virus, which was 2.10 mmol. L⁻¹. Likewise, it was observed that the

genotype GM-3034 had the highest level of magnesium (2.85 mmol. L⁻¹).

Table 3. The Effect of t CMV on the Content of Genotypes of Magnesium

Genotype	Magnesium		Mean
	Non-infected plants	Infected plants	
AH-2018	3.02	2.12	2.57
HA-144	2.53	1.29	1.91
HA-265	2.84	2.65	2.75
HA-17NM-106	2.73	1.89	2.31
HA-609	2.10	1.25	1.67
GA-1534	3.18	2.67	2.93
HA-2011	2.76	2.32	2.54
HA-17ME-214	3.19	2.32	2.76
GM-3034	3.60	2.85	3.22
GEDIZ F1	3.08	2.15	2.62
Mean	2.90	2.15	
L.S.D. (0.05)	Genotypes	Infection	Interaction
	0.61	0.27	0.86

Sodium

It was found that the viral infection had a significant effect in increasing the sodium concentration in the plant with a rate of 169.5 ppm compared with its concentration in uninfected plants, whose sodium content reached 149.67 ppm. Additionally, there was a noticeable difference between the genotypes' sodium contents, which ranged from genotype HA-609 HA-609 (178.22 ppm) to 137.75 ppm in the genotype GM-3034 . On the other hand, the sodium concentrations in the other genotypes varied from 178.22 to 137.75 ppm. (Table 4).

When infected and uninfected genotypes were compared, it was found that the infected GM-3034 genotype had the lowest sodium content (153.17 ppm), which was significantly lower than the sodium content of uninfected plants (122.33 ppm). It was also revealed that the infected GM-3034 genotype had the lowest sodium content. The genotype HA-609 had the greatest sodium content (186.90 ppm), which was much higher than the 169.53 ppm found in plants that were uninfected.

Table 4. The Effect of CMV on the Genotype Content of Sodium

Genotype	Sodium (ppm)		Mean
	Non-infected plants	Infected plants	
AH-2018	157.13	164.93	161.03
HA-144	160.73	182.07	171.40
HA-265	136.70	155.67	146.18
HA-17NM-106	175.80	178.63	177.22
HA-609	169.53	186.90	178.22
GA-1534	133.43	162.37	147.90
HA-2011	136.17	165.83	151.00
HA-17ME-214	158.13	170.90	164.52
GM-3034	122.33	153.17	137.75
GEDIZ F1	146.70	174.87	160.78
Mean	149.67	169.53	
L.S.D. (0.05)	Genotypes	Infection	Interaction
	2.81	1.25	3.97

Potassium

The findings showed that there was an unusual increase in potassium content in CMV-infected plants, with a rate of 142.42 ppm, which was significantly higher than the concentration of 128.43 ppm found in uninfected plants. The potassium level of the genotypes differed among them as well; the HA-609 genotype had the highest value (135.45 ppm), while the GA-1534

genotype had the lowest potassium content (132.15 ppm). The calcium level of the remaining genotypes ranged between 135.45 and 132.15 ppm (Table 5). The interaction between genotypes and CMV infection demonstrated that the infected HA-609 genotype had the highest potassium content (152.33 ppm), which was significantly higher than the potassium content in plants that were not infected with the virus, which was equal to 118.57 ppm.

Table 5. The Effect of CMV on Potassium Content of the Melon Genotypes

Genotype	Potassium		Mean
	Non-infected plants	Infected plants	
AH-2018	*142.00	149.33	145.67
HA-144	134.87	163.97	149.42
HA-265	120.23	135.60	127.92
HA-17NM-106	131.60	137.27	134.43
HA-609	118.57	152.33	135.45
GA-1534	130.50	133.80	132.15
HA-2011	138.60	144.50	141.55
HA-17ME-214	121.20	137.20	129.20
GM-3034	121.27	135.10	128.18
GEDIZ F1	125.50	135.10	130.30
Mean	128.43	142.42	
L.S.D _(0.05)	Genotypes	Infection	Interaction
	2.97	1.33	4.21

According to Mathews (1981), viral infections clearly affect a plant's content of several major and minor mineral elements to differing levels, which in turn causes a decrease in plant weakness and its reflection on various growth parameters. These findings confirmed those of Jones and Huber (2012), who founded that most physiological processes are impacted when plants become infected with diseases, including viral diseases like infection with the *Tobacco Mosaic Virus* (TMV), which has a direct impact on the decrease in the mineral content of the plant, specifically calcium, magnesium, and manganese. It was also demonstrated in a study carried out by Ilyas et al. (2016) that infection with TYLCV caused a decrease in the calcium, magnesium, and manganese content of the tomato plant as well as a significant reduction in the amount and sizes of cells.

Al-Tamimi (2019) demonstrated that TYLCV infection caused the plant's content of several elements, such as calcium, magnesium, and manganese, to significantly decrease.

The current study's findings also concurred with those reported by Wang et al., (2013) and Shabala and Pottosin (2014), who both stated that when plants are exposed to biotic and non-biotic stresses, they exhibit abnormal responses in an effort to withstand those stresses, including increased absorption of some mineral elements like sodium and potassium. And at a level greater than the plant's typical concentration of these components. In a previous study, it was found that the TYLCV infection caused a significant increase in sodium and potassium levels in infected plants as a result of the stress that the virus causes, which causes an increase in osmotic pressure in plant cells and increases water

absorption as well as some dissolved elements in the soil (Khalil et al., 2014).

Effect of CMV on Some Hormones in Fruits of Melon Genotypes

Gibberellin

The results given in Table (6) clearly show that the plants that were not infected with the virus were significantly better than the infected plants due to their gibberellin concentration, which was 4.96 mmol/ml compared to 2.85 mmol/ml for the infected plants. The gibberellin levels among the melon plant genotypes also differed significantly. The GA-1534 genotype was clearly

superior to the other genotypes, having the greatest rate of the hormone, 5.37 mmol/ml, as compared to the HA-609 genotype, which had a rate of 2.90 mmol/ml. It was also found that the genotype HA-609 infected with CMV was most affected in reducing the plant gibberellin content (1.61 mmol/ml) with a significant difference from the content of the same hormone in plants of the same uninfected genotype, which amounted to 4.20 mmol/ml. While the GA-1534 genotype had the least reduction in the gibberellin content of the plant, which amounted to 4.36 mmol/ml, with a significant difference from the rate of its presence in uninfected plants of the same genotype (6.38 mmol/ml).

Table 6. The Effect of CMV on the Plant Content of Gibberellin Hormone in Melon Genotypes

Genotype	Gibberellin (mmol/ ml)		Mean
	Non-infected plants	Infected plants	
AH-2018	5.02	1.92	3.47
HA-144	5.38	1.50	3.44
HA-265	3.44	2.58	3.01
HA-17NM-106	4.32	2.78	3.55
HA-609	4.20	1.61	2.90
GA-1534	6.38	4.36	5.37
HA-2011	5.18	3.59	4.39
HA-17ME-214	4.91	3.22	4.06
GM-3034	5.74	3.65	4.69
GEDIZ F1	5.02	3.28	4.15
Mean	4.96	2.85	
L.S.D _(0.05)	Genotypes	Infection	Interaction
	0.60	0.27	0.86

Cytokinin

The findings demonstrated that CMV infection increased the cytokinin concentration to 4.11 mmol/ml, a considerable increase over the unaffected genotypes' average cytokinin concentration of 2.39 mmol/ml. The impacted genotypes also had hereditary factors in the level of the plant cytokinin hormone, with genotype HA-609 having the highest amount (3.80 mmol/ml), which was significantly higher than that of other genotypes (Table 7). The cytokinin

hormone concentration was lowest in the AH-2018 genotype, with an average of 2.90 mmol/ml. The genotype HA-609 infected with the virus had the highest concentration of cytokinin hormone, which recorded an average of 6.02 mmol/ml. This was much higher than the content of uninfected plants, which was 1.57 mmol/ml, while the genotype given AH-2018 had the lowest value. Compared to plants not infected with the virus, which had a cytokinin concentration of 2.57 mmol/ml, the lowest amount was 3.24 mmol/ml.

Table 7. The effect of CMV on the Cytokinin Content of Melon Genotypes

Genotype	Cytokinin (mmol/ ml)		Mean
	Non-infected plants	Infected plants	
AH-2018	2.57	3.24	2.90
HA-144	1.56	5.58	3.57
HA-265	2.76	3.45	3.11
HA-17NM-106	2.28	4.08	3.18
HA-609	1.57	6.02	3.80
GA-1534	2.31	3.28	2.79
HA-2011	2.83	3.76	3.30
HA-17ME-214	3.16	4.53	3.85
GM-3034	2.44	3.67	3.06
GEDIZ F1	2.46	3.50	2.98
Mean	2.39	4.11	
L.S.D _(0.05)	Genotypes	Infection	Interaction
	0.51	0.23	0.73

Conclusions

The genetic differences of the melon plants used in this study, as well as their varied responses to viral infection and the effect of the virus on the internal contents of the plant cells and its negative effects on the vital functions of the plant and the construction of materials required for the perpetuation of plant life, including the level of hormone overgrowth, may all be contributing factors to the variation in the content of the hormone gibberellin in these plants. Researchers found that the TYLCV infection of the tomato plant resulted in a notable decrease in the amount of the hormone gibberellin in the plant, which had an adverse effect on the division and elongation of the cells of the affected plant and the appearance of the plant with various disease symptoms, such as plant stunting, short internodes, and Leaf curl (Karim, 2016). Giron and Glevarec (2014) reported an increase in the levels of some plant hormones, including cytokinin, in some plants infected with some pathogens, including infection with some plant viruses such as Cabbage leaf curl virus. Richard et al. (2014) indicated that increasing the concentration of indole acetic acid (IAA) in infected plants is a defense against pathogen action.

References

- AL-Abedy, A. N., Al-Fadhal, F. A., Karem, M. H., Al-Masoudi, Z., & Al-Mamoori, S. A. (2018). Genetic variability of different isolates of *Rhizoctonia solani* Kühn isolated from Iranian imported potato tubers (*Solanum tuberosum* L.). *International Journal of Agriculture and Statistical Sciences*, 14(2), 587-598.
- Al-Abedy, A. N., Kadhim, J. H., Abdalmoohsin, R. G., & Al-Taey, D. K. (2021). Genetic diversity of tomato yellow leaf curl virus isolates and the effect of virus on the hormones content of tomato (*Solanum lycopersicum*) plants. *Research on Crops*, 22(2). <http://dx.doi.org/10.31830/2348-7542.2021.078>
- Al-Fadhal, F. A., AL-Abedy, A. N., & Alkhafije, D. A. (2019). Isolation and molecular identification of *Rhizoctonia solani* and *Fusarium solani* isolated from cucumber (*Cucumis sativus* L.) and their control feasibility by *Pseudomonas fluorescens* and *Bacillus subtilis*. *Egyptian Journal of Biological Pest Control*, 29(1), 1-11. <http://dx.doi.org/10.1186/s41938-019-0145-5>
- Al-Fadhal, F.A.; A.N. AL-Abedy and M.M. Al-Janabi (2018). Molecular identification of novel isolates of *Rhizoctonia solani* Kühn and *Fusarium* spp. (Matsushima) isolated from

petunia plants (*Petunia hybrida* L.). *Plant Archives*, 18 (1), 703-711.

Al-Fahham, A.A. (2018) Development of New LSD Formula when Unequal Observations Numbers of Observations Are. *Open Journal of Statistics*, 8, 258-263. <https://doi.org/10.4236/ojs.2018.82016>

Al-Salami, I., Al-Gwaree, R. N., & Al-Abedy, A. N. (2019). Genetic Relationship among Some Isolates of *Rhizoctonia solani* Isolated from Some Infected Tomato Plants (*Solanum lycopersicum* L.). *Journal of Global Pharma Technology*, 11(7), 379-385.

Al-Yasiri, H.I.A. (2016). *The use of molecular and immunological methods and plant reagents in the diagnosis of cucumber mosaic virus CMV and induction of resistance in cucumber plants using phyto-stimulating bacteria*. University of Kufa, College of Agriculture, Plant Protection Department.

Asad, Z., Ashfaq, M., & Ahsan, M. (2019). Serological and Molecular Identification Based on Coat Protein (CP) Gene of Cucumber mosaic virus (CMV) Infecting Cucumber (*Cucumis sativus* L) in Pothwar Region of Pakistan. *J Plant Biochem Physiol*, 7(231), 2. <http://dx.doi.org/10.4172/2329-9029.1000231>

Chen, Y. K. (2003). *Occurrence of cucumber mosaic virus in ornamental plants and perspectives of transgenic control*. Wageningen University and Research.

Chitara, M. K., Maurya, S. K., Singh, R. P., & Srivastava, J. N. (2022). Major Diseases of Leafy Vegetables and Their Management. In *Diseases of Horticultural Crops* (pp. 283-323). Apple Academic Press.

El-Gendi, H., Al-Askar, A. A., Király, L., Samy, M. A., Moawad, H., & Abdelkhalek, A. (2022). Foliar Applications of *Bacillus subtilis* HA1 Culture Filtrate Enhance Tomato Growth and Induce Systemic Resistance against Tobacco mosaic virus Infection. *Horticulturae*, 8(4), 301. <https://doi.org/10.3390/horticulturae8040301>

Faostat Database (2015) watermelon world production statistics. Retrieved from http://www.growtomatoes.com/world_productionstatistics.html

Giron, D. and Glevarec, G. (2014). Cytokinin-induced phenotypes in plant-insect interactions: learning from the bacterial world. *Journal of Chemical Ecology*, 8-11. <https://doi.org/10.1007/s10886-014-0466-5>

Hull, R. (2002). *Matthew's Plant Virology*. 4th ed. Academic Press, New York, NY. 1001.

Ilyas, M., Ahmad, M., Hussain, Z., Saeed, M., Luqman, F.B., Shah, K., Mohammad Khan, I. & Shah, S. (2016). Interactive effect of calcium and magnesium on the growth and yield of tomato (*Lycopersicon esculentum* L.). *Pure Applied Biology*, 5(4), 876-882. <https://doi.org/10.19045/BSPAB.2016.50110>

Jones, J.I. (1984). In official methods of analysis of the association of official analytical chemists. Ed. Williams, S. *Association of official analytical chemists*. Arlington, Virginia. USA. 38-64.

Karim, M.H. (2016). *Immunological and molecular diagnosis of tomato yellow leaf curl virus and evaluation of the response of some tomato cultivars to infection*. Master Thesis. faculty of Agriculture. Al-Muthanna University. 104 pages.

Kimura, S., Ohkawara, H., Minakawa, K., Fukatsu, M., Mori, H., Takahashi, H., ... & Ikezoe, T. (2020). Optimal timing of apheresis for the efficient mobilization of peripheral blood progenitor cells recruited by high-dose granulocyte colony-stimulating factor in healthy donors. *Transfusion and Apheresis Science*, 59(3), 102737. <https://doi.org/10.1016/j.transci.2020.102737>

Manchali, S., Chidambara Murthy, K. N., & Patil, B. S. (2021). Nutritional composition and health benefits of various botanical types of melon (*Cucumis melo* L.). *Plants*, 10(9), 1755. <https://doi.org/10.3390/plants10091755>

Mathews, R.E.F. (1981). *Plant Virology*. 2nd ed. Academic Press New York.

Murphy, C. A., Langrish, C. L., Chen, Y., Blumenschein, W., McClanahan, T., Kastelein, R. A., ... & Cua, D. J. (2003). Divergent pro-and antiinflammatory roles for IL-23 and IL-12 in joint autoimmune inflammation. *The Journal of experimental medicine*, 198(12), 1951-1957. <https://doi.org/10.1084/jem.20030896>

Ortali, A., & Onarinde, B. (2021). Growth Potential of *Listeria monocytogenes* on Cantaloupe (*cucumis melo*) pulp evaluation under a dynamic time-temperature profile. Retrieved from <https://eprints.lincoln.ac.uk/id/eprint/46086/1/iPoster%20Gallery%20-%20Cantaloupe.pdf>

Page, A.L., Miller, R.H. & Keeney, D. R. (1982). *Method of soil and analysis Part 2, 2nd ed, Agron.9*. Publisher, Madison, Wisconsin, USA.

Richard, O., Sue, M., Musser, H., Gallucci, M., Rochers, B.D. & Brown, J.K. (2014). Microarray analysis of tomato plants exposed to the nonviruliferous or viruliferous whitefly

vector harboring *Pepper golden mosaic virus*. *Journal of Insect Science*, 1-10.

Shabala, S. & Pottosin, I. (2014). Regulation of potassium transport in plants under hostile conditions: implications for abiotic and biotic stress tolerance. *Physiologia Plantarum an International Journal for Plant Biology*, 257-279. <https://doi.org/10.1111/ppl.12165>

Torrance, L., Cowan, G. H., McLean, K., MacFarlane, S., Al-Abedy, A. N., Armstrong, M., ... & Bryan, G. J. (2020). Natural resistance to Potato virus Y in *Solanum tuberosum* Group Phureja. *Theoretical and Applied Genetics*, 133(3), 967-980. <https://doi.org/10.1007/s00122-019-03521-y>

Tungadi, T., Watt, L. G., Groen, S. C., Murphy, A. M., Du, Z., Pate, A. E., ... & Carr, J. P. (2021). Infection of *Arabidopsis* by cucumber mosaic virus triggers jasmonate-dependent resistance to aphids that relies partly on the pattern-triggered immunity factor BAK1. *Molecular Plant Pathology*, 22(9), 1082-1091. <https://doi.org/10.1111/mpp.13098>